First Aid Basics

Knowledge document 046 of 100

Introduction to First Aid

First aid is the immediate help given to a person who is injured or suddenly ill, before professional medical help arrives. Its three main aims, often remembered as the three Ps, are to preserve life, prevent the condition from getting worse and promote recovery. Knowing basic first aid can save lives, because in emergencies such as cardiac arrest, severe bleeding or choking, every minute counts. First aid is not a substitute for medical treatment: after giving first aid, the person should be taken to a doctor or hospital if the problem is serious. Organisations such as the Indian Red Cross Society, founded in 1920, and the St. John Ambulance train people in first aid, and many schools and colleges include it in their programmes.

Emergency Number in India

The single emergency number in India is 112. It is the integrated Emergency Response Support System (ERSS) launched across the country in 2019 and can be dialled from any phone to reach police, fire and ambulance services. Older numbers still work in many places: 100 for police, 101 for fire, 102 for an ambulance for pregnant women and children, 108 for emergency ambulance services, 1091 for women's helpline and 1098 for the child helpline. In Tamil Nadu, the 108 ambulance service, started in 2008, is widely used and attends emergencies free of charge, and the 112 app and the Kaaval Udhavi app can also send alerts. When calling for help, stay calm, state the exact location and the nature of the emergency, the number of people hurt, and do not hang up until told to do so.

First Steps in Any Emergency

First check for danger to yourself, the casualty and bystanders, such as traffic, fire or electricity, and make the area safe before approaching. Then check whether the person responds by tapping the shoulders and shouting. Call for help or dial 112. Check the airway and breathing. A commonly taught sequence is DRSABCD: Danger, Response, Send for help, Airway, Breathing, CPR and Defibrillation. Keep the person warm and calm and do not give food or drink to an unconscious person.

What Are the Steps of CPR?

CPR, or cardiopulmonary resuscitation, is an emergency technique used when a person is unresponsive and not breathing normally, as in cardiac arrest. It keeps blood and oxygen flowing to the brain until medical help arrives. The steps are as follows. First, make sure the scene is safe. Second, check for response by tapping and shouting. Third, call 112 or ask someone to call, and ask for an automated external defibrillator (AED). Fourth, check for breathing for no more than 10 seconds; gasping is not normal breathing. Fifth, begin chest compressions: place the person on their back on a firm surface, kneel beside them, put the heel of one hand on the centre of the chest, which is the lower half of the breastbone, place the other hand on top and interlock the fingers, keep the arms straight and press hard and fast. Compressions should be 5 to 6 centimetres deep for an adult, at a rate of 100 to 120 per minute, which is about the beat of the song "Stayin' Alive", allowing the chest to rise fully between presses. Sixth, if trained, after 30 compressions give 2 rescue breaths by tilting the head back, lifting the chin, pinching the nose and breathing into the mouth for about one second each, so that the chest rises. Repeat cycles of 30 compressions and 2 breaths. Seventh, if an AED arrives, switch it on, follow the voice prompts, attach the pads and deliver a shock if advised. Continue until the person starts breathing, medical help takes over or you are too exhausted to continue. Untrained people are encouraged to do hands-only CPR, which uses chest compressions alone. For children, use one hand and compress about one-third of the chest depth. Early CPR can double or triple the chances of survival.

How Should a Minor Burn Be Treated?

A minor burn, such as a small superficial burn that only reddens the skin, or a small scald, should be cooled immediately under cool running tap water for at least 10 to 20 minutes. This stops the burning process, reduces pain and swelling. Remove rings, watches and tight clothing near the burn before swelling starts, but do not remove clothes stuck to the skin. Do not apply ice, ice water, butter, oil, toothpaste or turmeric, as these can damage the skin or cause infection. Do not burst blisters. After cooling, cover the burn loosely with a clean, non-fluffy dressing or cling film and take a simple pain reliever if needed. Seek medical help if the burn is larger than the person's palm, deep, blistering extensively, on the face, hands, feet, genitals or over a joint, caused by chemicals or electricity, or if the person is a child or an elderly person. For chemical burns, rinse with plenty of water for 20 minutes.

Bleeding, Fractures and Choking

For severe bleeding, apply firm direct pressure on the wound with a clean cloth or sterile dressing and do not remove it, if blood soaks through add more layers and raise the injured limb above the heart if possible, and call for help. For a nosebleed, sit up, lean forward and pinch the soft part of the nose for 10 minutes. For a suspected fracture, keep the injured part still, support it with a splint or sling and do not try to straighten it. For a choking adult who cannot speak or breathe, give up to five back blows between the shoulder blades, followed by up to five abdominal thrusts (the Heimlich manoeuvre), and alternate until the object comes out or the person becomes unconscious, when CPR must begin. For fainting, lay the person down and raise their legs. For heat stroke, move the person to a cool place, remove excess clothes, cool the body with water and call an ambulance.

Snakebite and Other Common Emergencies

In India, snakebite is a significant emergency, and the "big four" venomous snakes are the cobra, the common krait, Russell's viper and the saw-scaled viper. The person should be kept calm and still, the bitten limb immobilised at about heart level, jewellery removed and the person taken quickly to a hospital with antivenom. Do not cut the wound, suck the venom, apply a tight tourniquet or use home remedies. For an electric shock, switch off the power before touching the person. For seizures, move hard objects away, cushion the head and do not put anything in the mouth. For poisoning, call for medical help and a poison helpline immediately. In an allergic reaction with difficulty in breathing, call an ambulance.

First Aid Box

A basic first aid box contains sterile gauze and bandages, adhesive plasters, an antiseptic solution, cotton, scissors, tweezers, a crepe bandage, disposable gloves, a thermometer, a burn dressing and a list of emergency numbers. It should be checked regularly for expired items.

First Aid in India and Chennai

The Good Samaritan law in India, backed by the Supreme Court in 2016 and the Motor Vehicles (Amendment) Act, 2019, protects people who help road accident victims from legal harassment. Tamil Nadu's Innuyir Kappom Thittam, started in December 2021, offers free treatment for the first 48 hours to road accident victims in government and empanelled private hospitals. The golden hour after a serious injury is critical.

Challenges and Future

Challenges include low public awareness, fear of legal trouble and delays in ambulance arrival. Mobile apps, AED placement in public places and school first aid training are improving outcomes.

Quick Facts

- The emergency number in India is 112; 108 is the ambulance service.
- CPR uses 30 chest compressions at 100 to 120 per minute followed by 2 rescue breaths.
- A minor burn should be cooled under cool running water for 10 to 20 minutes.
- Do not apply ice, butter or toothpaste on burns.
- The main aims of first aid are to preserve life, prevent worsening and promote recovery.